

TITLE PAGE

- **Problem Statement ID - 1**
- **Problem Statement Title-** Eco-Loop Building Agents: AI-Powered Autonomous Smart Building Optimization Challenge
- **Theme-** Clean & Green Technology (Smart Automation / Sustainable Energy)
- **PS Category-** Software
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Eco-Loop: A Self-Correcting Building That Negotiates Its Own Energy Use

Proposed Solution:

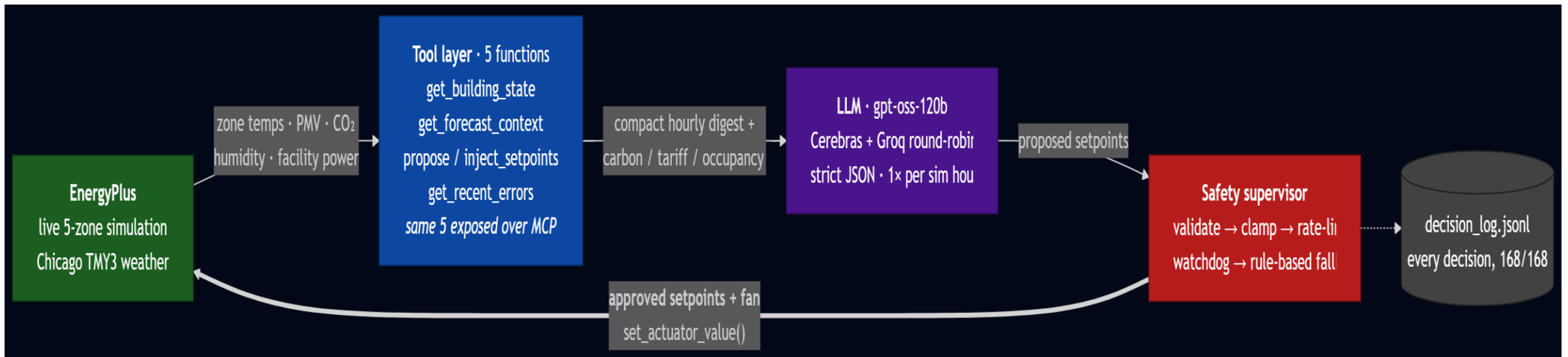
- **What it is:**
 - A closed control loop: **EnergyPlus** simulates a real 5-zone building; an **open-weight LLM** (gpt-oss-120b) reads live zone state every simulated hour and injects **new HVAC setpoints back into the running simulation**- one process, in-memory, no human in the loop.
 - The LLM doesn't just react to temperature. It reasons against a **24 h grid carbon-intensity + tariff forecast**, occupancy lookahead, weather lookahead, CO₂/IAQ, humidity, and a **peak-demand threshold** then pre-cools into clean, cheap hours and sets back hard when the building empties.
- **How it addresses the problem:**
 - Traditional BMS runs rigid schedules. Measured on a matched 7-simulated-day run, same weather, same building: **-8.8% total facility electricity, -23.3% HVAC-only electricity, and comfort improved 12.4 points (80.7% -> 93.1% of occupied hours with PMV in [-0.5, +0.5])**.
 - Savings are **not** taken out of occupants' comfort- the usual trade-off is inverted. Reheat gas also fell 32.9 -> 3.6 kWh and peak demand 19.9 -> 19.0 kW.
- **Innovation / uniqueness:**
 - **A safety supervisor wraps the LLM:** schema validation -> hard clamps -> anti-thrash rate limiter -> watchdog timeout -> deterministic rule-based fallback. The simulation cannot be killed by a bad, slow, or absent LLM response. A full run completes with the LLM entirely unreachable (shipped as proof).
 - **The LLM beats its own deterministic fallback on energy, HVAC energy, AND comfort simultaneously** so the intelligence is doing measurable work, not just adding risk.
 - **Genuine self-correction:** the agent was wasting 20+ kWh/day coasting into an empty building because the state digest described the previous hour. Adding one explicit decision_hour_occupied signal let the model fix itself- no supervisor override needed.

TECHNICAL APPROACH

Technologies Used:

- **Python 3.11; EnergyPlus 26.1** driven through its bundled pyenergyplus runtime API with in-process callbacks (not file-rewrite-and-rerun, not FMU/BCVTB)
- **gpt-oss-120b** (OpenAI open-weight), served OpenAI-compatible on **Cerebras+Groq**, round robined per call; local **Ollama** documented as a fully offline option
- **FastMCP**: MCP server exposing the same 5 tools; **Pydantic** for strict-JSON decision validation; **Plotly** for the self-contained offline dashboard; **pandas/numpy** for metrics
- **Building model**: shipped **5ZoneAirCooled.idf**, Chicago TMY3 weather, 15-21 July

Methodology:



FEASIBILITY AND VIABILITY

- **Feasibility (Measured, not projected):**
 - Runs today on free-tier infrastructure. **168/168 hourly decisions injected, 0 controller errors**, 165/168 genuine LLM calls, latency **p50 816ms / p95 2120ms** against a 15s watchdog ~7x headroom.
 - **Reliability soak: 14 simulated days (2 × the required horizon) with the LLM live throughout- 336/336 decisions injected, 0/336 fell back to the rule schedule, 0 crashes**, latency p50 693ms / p95 1641ms, and 92.9% occupied comfort held across the full 14 days.
 - Energy figures come from **EnergyPlus's own meter file** (eplusmtr.csv), not self-reported totals, so the headline can't be doubted.
- **Challenges and risks:**
 - **~62% of facility electricity is lighting/plug load** no setpoint can touch which caps the achievable total-kWh%
 - Free-tier LLM rate limits: an early run had only 43% genuine LLM participation
 - LLM could return invalid, slow, or no output mid-run
 - Naive energy accounting silently over-counted by 16.5%
- **Strategies that resolved each:**
 - Report **HVAC-only % as the honest headline** alongside total- state the ceiling rather than hide it
 - Measured the actual binding limit per provider (they invert: Groq token-bound, Cerebras request-bound) -> per-provider throttles + round-robin took participation **43% -> 98%**.
 - The safety supervisor: clamp + watchdog + deterministic fallback. A **100%-fallback run** is shipped as proof the loop survives a total LLM outage
 - Caught it by reconciling against E+'s own meters; the smoke test now asserts **agreement within 0.5%** as a permanent regression guard (reconciles to 0.00%)
 - Method, not luck: **Two optimizations (CO₂-gated fan-off, optimal-start pre-heat)** were built, measured, and reverted when the data showed **no net benefit**- negative results documented, not buried.

ARTIFACTS

- **Full source:** <https://github.com/mahendra-kausik/eco-loop-building-agents>
- **Architecture doc:** docs/ARCHITECTURE.md: tool-calling, prompt strategy, latency management, long-log handling
- **Dashboard:** results/dashboard.html
- **Building models:** models/baseline.idf + agent_llm_runtime.idf / agent_fallback_runtime.idf (the actual applied hour-by-hour schedules baked into real IDF objects, diffable against baseline)
- **Decision log:** results/decision_log.jsonl (168 decisions with the model's own stated reasoning) + decision_log_llm_unreachable.jsonl (the 100%-fallback proof run)
- **Reliability soak log:** results/decision_log_soak14.jsonl (336 decisions, 14 simulated days, 0 fallbacks) plus the soak strip on results/dashboard.html

Metric	Baseline	AI closed loop	Result
Total electricity	1100.1 kWh	1003.6 kWh	−8.8%
HVAC-only electricity	413.7 kWh	317.3 kWh	−23.3%
Comfort (PMV in band, occupied)	80.7%	93.1%	+12.4 pts
Reheat gas	32.9 kWh	3.6 kWh	−89%
Peak demand	19.9 kW	19.0 kW	−0.9 kW
CO ₂ (grid-weighted)	663.8 kg	602.2 kg	−61.6 kg
Cost @ ₹6/kWh	₹6079	₹5485	−₹594
Crash-free horizon	-	14 sim days	336/336, 0 fallback

RESEARCH AND REFERENCES

- **EnergyPlus** documentation & Python Plugin/API guide: energyplus.net (US DOE)
- **ASHRAE Standard 55 / Fanger PMV** thermal comfort model: the basis of the $[-0.5, +0.5]$ comfort band
- **Model Context Protocol (MCP)** specification: modelcontextprotocol.io
- **gpt-oss-120b** (OpenAI open-weight model), served via **Cerebras** and **Groq** OpenAI-compatible APIs
- **ASHRAE Standard 62.1**: ventilation / CO₂ generation defaults used for the IAQ tracer